

Substring Order II

Time limit: 1.00 s

Memory limit: 512 MB

You are given a string of length n . If all of its substrings (not necessarily distinct) are ordered lexicographically, what is the k th smallest of them?

Input

The first input line has a string of length n that consists of characters a–z.

The second input line has an integer k .

Output

Print the k th smallest substring in lexicographical order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq \frac{n(n+1)}{2}$

Example

Input	Output
baabaa 10	ab

Explanation: The 10 smallest substrings in order are a, a, a, a, aa, aa, aab, aaba, aabaa, and ab.